

Coordinate Systems and Change of Basis

Lecture Notes

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Introduction

In linear algebra, we often express vectors using coordinates, like $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. Implicitly, we are using the **standard basis** of the vector space. For $\mathbb{R}^3$, the standard basis $\mathcal{S}$ is $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ where:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

So, $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is shorthand for $\mathbf{x} = 1\mathbf{e}_1 + 2\mathbf{e}_2 + 3\mathbf{e}_3$.

However, it is often useful to use a different **non-standard basis**. This lecture covers how to translate vector representations between different coordinate systems. We will explore:

1. How to find a vector's standard coordinates when given its representation in a different basis $\mathcal{B}$.
2. How to find a vector's representation in a basis $\mathcal{B}$ when given its standard coordinates.
3. The powerful **change of coordinates matrix**, a tool that automates these conversions.

1 From a Custom Basis to the Standard Basis

💡 Key Idea and Notation

Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$ be a basis for a vector space V . Any vector $\mathbf{x}$ in V can be written as a unique linear combination of these basis vectors:

$$\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_n\mathbf{b}_n$$

The scalars $(c_1, c_2, \dots, c_n)$ are the **coordinates of $\mathbf{x}$ relative to the basis $\mathcal{B}$** . We denote this coordinate vector as $[\mathbf{x}]_{\mathcal{B}}$.

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Converting from the basis $\mathcal{B}$ to the standard basis is simply a matter of performing this linear combination.

1.1 Example 1: Find the standard coordinates of a vector

Problem: We are working in $\mathbb{R}^3$ with the basis $\mathcal{B}$ given by:

$$\mathbf{b}_1 = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix}, \quad \mathbf{b}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

A vector $\mathbf{x}$ has coordinates $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ relative to this basis. Find the vector $\mathbf{x}$ in the standard coordinate system.

Solution (Step-by-Step)

1. **Understand the notation:** $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$ tells us the weights for the linear combination.

$$\mathbf{x} = 2\mathbf{b}_1 + 1\mathbf{b}_2 + (-3)\mathbf{b}_3$$

2. **Substitute the basis vectors:**

$$\mathbf{x} = 2 \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} + 1 \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix} - 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

3. **Perform scalar multiplication and vector addition:**

$$\mathbf{x} = \begin{bmatrix} 4 \\ 4 \\ 8 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix} + \begin{bmatrix} -3 \\ -3 \\ -3 \end{bmatrix} = \begin{bmatrix} 4 - 1 - 3 \\ 4 + 0 - 3 \\ 8 + 3 - 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 8 \end{bmatrix}$$

Conclusion: The vector $\mathbf{x}$ in the standard basis is $\begin{bmatrix} 0 \\ 1 \\ 8 \end{bmatrix}$.

2 From the Standard Basis to a Custom Basis

💡 Key Idea

This is the reverse problem: given $\mathbf{x}$ in standard coordinates, find $[\mathbf{x}]_{\mathcal{B}}$. We need to find scalars $c_1, \dots, c_n$ such that:

$$c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \dots + c_n\mathbf{b}_n = \mathbf{x}$$

This vector equation can be rewritten as a matrix equation. Let $P_{\mathcal{B}}$ be the matrix whose columns are the basis vectors of $\mathcal{B}$. Then the equation is:

$$P_{\mathcal{B}} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \mathbf{x}$$

The vector we are solving for, $[c_1, \dots, c_n]^T$, is precisely $[\mathbf{x}]_{\mathcal{B}}$.

2.1 Example 2: Find the coordinates relative to a basis $\mathcal{B}$

Problem: Use the same basis $\mathcal{B}$ from Example 1. The vector $\mathbf{x}$ in standard coordinates is $\mathbf{x} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$. Find $[\mathbf{x}]_{\mathcal{B}}$.

Solution (Step-by-Step)

1. **Set up the vector equation:** We need to find c_1, c_2, c_3 such that:

$$c_1 \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 3 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

2. **Convert to an augmented matrix** and solve the system $P_{\mathcal{B}}\mathbf{c} = \mathbf{x}$:

$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 2 \\ 2 & 0 & 1 & 3 \\ 4 & 3 & 1 & 4 \end{array} \right]$$

3. **Solve using row reduction:** The reduced row echelon form is:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 5 \end{array} \right]$$

4. **Interpret the solution:** This gives $c_1 = -1$, $c_2 = 1$, and $c_3 = 5$.

Conclusion: The coordinates of $\mathbf{x}$ relative to basis $\mathcal{B}$ are $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$.

3 The Change of Coordinates Matrix

A **change of coordinates matrix**, P , is a matrix that performs coordinate conversion via a simple matrix multiplication.

3.1 Changing from Basis $\mathcal{B}$ to the Standard Basis $\mathcal{S}$

The matrix that converts $[\mathbf{x}]_{\mathcal{B}}$ to $\mathbf{x}$ is denoted $P_{\mathcal{S} \leftarrow \mathcal{B}}$.

$$\mathbf{x} = P_{\mathcal{S} \leftarrow \mathcal{B}} \cdot [\mathbf{x}]_{\mathcal{B}}$$

💡 Construction: B to Standard Basis (S)

The matrix $P_{\mathcal{S} \leftarrow \mathcal{B}}$ is simply the matrix whose columns are the basis vectors of $\mathcal{B}$, written in standard coordinates.

$$P_{\mathcal{S} \leftarrow \mathcal{B}} = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_n]$$

3.2 Changing Between Two Non-Standard Bases ($P_{\mathcal{A} \leftarrow \mathcal{B}}$)

The matrix that converts $[\mathbf{x}]_{\mathcal{B}}$ into $[\mathbf{x}]_{\mathcal{A}}$ is $P_{\mathcal{A} \leftarrow \mathcal{B}}$.

$$[\mathbf{x}]_{\mathcal{A}} = P_{\mathcal{A} \leftarrow \mathcal{B}} \cdot [\mathbf{x}]_{\mathcal{B}}$$

💡 Construction: General Case (B to A)

The columns of $P_{\mathcal{A} \leftarrow \mathcal{B}}$ are the $\mathcal{B}$ -basis vectors, but written in $\mathcal{A}$ -coordinates.

$$P_{\mathcal{A} \leftarrow \mathcal{B}} = [\mathbf{b}_1]_{\mathcal{A}} \quad [\mathbf{b}_2]_{\mathcal{A}} \quad \cdots \quad [\mathbf{b}_n]_{\mathcal{A}}$$

To find each column $[\mathbf{b}_i]_{\mathcal{A}}$, we must solve the system $P_{\mathcal{A}} \mathbf{c} = \mathbf{b}_i$.

Example 4: Change between two non-standard bases

Problem: Given two bases for $\mathbb{R}^3$:

$$\mathcal{A} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} \right\}, \quad \mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \right\}$$

Find the change of coordinates matrix $P_{\mathcal{A} \leftarrow \mathcal{B}}$.

Solution

We need to find $[\mathbf{b}_1]_{\mathcal{A}}$, $[\mathbf{b}_2]_{\mathcal{A}}$, and $[\mathbf{b}_3]_{\mathcal{A}}$ by solving three separate linear systems. This can be done efficiently by creating one large augmented matrix $[P_{\mathcal{A}} \mid P_{\mathcal{B}}]$ and row reducing it to $[I \mid P_{\mathcal{A} \leftarrow \mathcal{B}}]$.

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 2 & 3 & 0 & -1 \\ 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 0 & 3 & 1 & 1 & 0 \end{array} \right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{2}{3} & 1 & -3 \\ 0 & 1 & 0 & -\frac{1}{3} & -2 & 0 \\ 0 & 0 & 1 & \frac{1}{3} & 1 & 1 \end{array} \right]$$

Conclusion: The change of coordinates matrix is $P_{\mathcal{A} \leftarrow \mathcal{B}} = \begin{bmatrix} 2/3 & 1 & -3 \\ -1/3 & -2 & 0 \\ 1/3 & 1 & 1 \end{bmatrix}$.

4 Summary & Applications

Practical Applications in Data Science & ML

- **Principal Component Analysis (PCA):** PCA finds a new basis for data where the new basis vectors (principal components) are orthogonal and aligned with directions of maximum variance. The change of coordinates matrix projects the data into this new, more informative system for dimensionality reduction.
- **Image Compression (JPEG):** The JPEG algorithm performs a change of basis (using the Discrete Cosine Transform) from a pixel basis to a frequency basis. In this new coordinate system, it's easy to discard high-frequency coordinates that the human eye is less sensitive to, thereby compressing the image.
- **Support Vector Machines (SVMs) with Kernels:** The "kernel trick" can be viewed as an implicit change of basis, mapping data into a higher-dimensional space where a linear separator can be found without explicitly forming the coordinates in that new space.

5 Exercise Problems

1. **From Basis to Standard:** Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right\}$ be a basis for $\mathbb{R}^2$. If a vector $\mathbf{x}$ has coordinates $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, what is $\mathbf{x}$ in the standard basis?
2. **From Standard to Basis:** Using the same basis $\mathcal{B}$ from question 1, find the coordinates $[\mathbf{y}]_{\mathcal{B}}$ for the standard vector $\mathbf{y} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$.
3. **Change of Coordinates Matrix to Standard:** Let $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ be a basis for $\mathbb{R}^3$. Find the change of coordinates matrix $P_{\mathcal{S} \leftarrow \mathcal{B}}$.
4. **Using the Matrix:** Use the matrix you found in question 3 to find the standard vector $\mathbf{x}$ whose $\mathcal{B}$ -coordinates are $[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$.